

MAD-1 PROJECT REPORT: TREKKING MANAGEMENT APP V1

1. STUDENT DETAILS

Roll Number: 24011P0566

Email: 24f2002823@ds.study.iitm.ac.in

Student Bio: Student pursuing B.Tech + M.Tech Dual Degree (IDP) in Computer Science and Engineering at JNTUH.

2. PROJECT DESCRIPTION & STATEMENT

The Trekking Management Application V1 is a multi-role web platform built to streamline trek operations and bookings.

Key Roles & Features:

- **Admin:** Pre-configured superuser managing users, trek staff approvals, trek destination creation, and staff assignments.
- **Trek Staff:** Self-registered users capable of managing assigned treks, tracking registered trekkers, and updating trek statuses.
- **User (Trekker):** Self-registered users able to search, filter, book open treks, and view their booking history.

3. AI / LLM DECLARATION

Primary AI/LLM Tool Used: Gemini 3.6 Flash / Google Antigravity AI Agent

Estimated Usage Percentage: ~15%

Declaration Details: An AI Agent (Gemini 3.6 Flash) was utilized via Google Antigravity IDE for initial setup, drafting SQLAlchemy models, generating Jinja2 Bootstrap HTML boilerplate templates, and debugging route interactions. All core business logic, user authorization rules, database constraints, and final end-to-end testing were verified and implemented by the student.

4. TECHNOLOGIES USED & DATABASE SCHEMA

Backend Framework: Flask (Python)

Frontend Template Engine: Jinja2 + HTML5 + Bootstrap 5 (CSS)

Database: SQLite via Flask-SQLAlchemy

Authentication: Flask-Login / Session Management with Passlib/Werkzeug hashing

Database Tables & Relationships:

1. User (id, username, email, password_hash, role [admin, staff, user], status [pending, approved, blacklisted])
2. Trek (id, title, description, location, price, capacity, status [open, closed, completed], created_at)
3. Booking (id, user_id [FK->User.id], trek_id [FK->Trek.id], booking_date, status [confirmed, cancelled])
4. StaffAssignment (id, staff_id [FK->User.id], trek_id [FK->Trek.id])

5. REST API ENDPOINTS

- **GET /api/treks** : List all active/open treks
- **GET /api/treks/<id>** : Retrieve specific trek details
- **POST /api/bookings** : Book a trek slot (authenticated user)
- **GET /api/user/bookings** : Fetch booking history for logged-in user

6. VIDEO PRESENTATION LINK

Demo Video URL:

https://drive.google.com/file/d/1ETGjVZUepJOi-_TQLDzbfjvQB2zC8aS/view?usp=sharing

(Access setting: Anyone with the link can view)